

Data Modeling Unit 1 Notes

DEPARTMENT OF COMPUTER ENGINEERING - ACADEMIC YEAR 2025-2026

Subject: Data Visualization and Modeling | Unit 1: Foundations of Statistical Data Modeling

Topics: Sampling Distributions, Central Limit Theorem, Probability Density, Normal Approximation.

1. Introduction to the Central Limit Theorem (CLT):

The Central Limit Theorem states that the sampling distribution of the sample mean approaches a normal distribution as the sample size increases, regardless of the shape of the population distribution. Specifically, given independent and identically distributed (i.i.d.) random variables with mean μ and finite variance σ^2 , the sample mean is approximately distributed as $N(\mu, \sigma^2 / n)$.

2. Practical Significance in Data Modeling:

The Central Limit Theorem is fundamental in inferential statistics, data visualization, and hypothesis testing because it permits calculating confidence intervals and p-values even when working with skewed data distributions in real-world visual analytics.

3. Visualizing Sampling Distributions:

When sampling from uniform or exponential distributions, visual plots demonstrate that histograms of the sample mean form a symmetric bell curve as sample size n exceeds 30.